7 Days

Training Performance, Optimization, LLM arxitektu- rasi (Final) va Tokenization.

1. LLM vs SLM. Statik ehtimollikdan → Ixcham tor doiradagi SLM va global emergent naohtiyga ega ulkan modelgacha LLM

2. Token vs Word: Söz Inson söz birligi. Token - BPE bölakchasi (subword)

3. Claude vs Local LLM: Bulutli Superkompyuter servislari va xususiy GPU dagi local modellar

4. Chinchilla & Scaling Law: D ≈ 20N qoidasi Compute optimal pretraining.

5. Embedding vs Encoding: Statik shart l özgaruvchi Encoding vs köp ölchamli semantic va dinamik vektorial koordinatalar.

6. Attention & layers: Q, V, K, orqali kontekst shakllantiruvchi Attention hamda faktual bilimlarni yuklachi FFN qatlamlari

7. Parallel vs Serial: Attention va FFN ning ketma ket Serial yoki Kernel Fusion orqali parallel hisoblanishi.

8. Layer Norm vs RMS Norm: Mean (μ) va shift (β) nichiga rib tashlab, GPU xotira otkazuvchanligini 10% 50% ga

Arxitektura codi tayyor bo'lgadan keyin eng masal-yatli eng qimmat bosqich. Train Phase.

Tez Arzon Xatoliklarsiz

3ta Asosiy Engineering Ustuniga tayanadi

LLM Training Performance.

↓

Hyperparameters	Training Stability	Regularization
LR	Exploding Grad	Overfitting
Batch Size	Loss = NaN/Inf	Weight Decay
Optimizer	LR Warm Up	Drop Out.
Warmp Up Steps	Grad Clipping	
Weight Decay		

Training Hyperparametrs

Hyperparametrs → bu train jarayonida avval engineer tomonidan qo'lda sozlanadigan model train tezligi va sifatini belgilovchi boshqaruv dastagidir.

LR → Har bir optimizer loss gradientga qarab Neyron vaznlarini nechog'lig katta mantiqiy qadam bilan o'zgartirishni belgilovchi ko'rsatkich

LR katta bo'lsa model konvergensiyaga qilmaydi

LR o'ta kichik bo'lsa model o'rganishi soatlar yillarga cho'zilib ketadi.

Batch size → Parallel ravishda GPU ga bir vaqtda yuklanadigan tokenlar va matnlar to'plami. Katta batch size GPU parallelizmidan maxt foydalanish imkonini beradi.

Optimizer (AdamW)
Gradientlarni hisoblab, Neyron vaznlarini yangilovchi matematik algoritm. LLM lardek birinchi va ikkinchi tartibli momentlarni (m+v^t) hisob lovchi AdamW.

Warm up Steps
Training eng boshida LR 0 dan belgilangan maksimal qiymatgacha sekin asta va ehtiyotkorlik bilan oshirib boradigan dastlabki iteratsiya soni (masalan dastlabki 2000 qadam)

Weight Decay → Neyron vaznlarini (W) shiddat bilan ösib ketishini oldini olish va ularning noldga qarab tortuvchi L2 penalt mexanizmi

Training Stability
LLM pre trained jarayoni yuzlab million dollar va oylar vaqt talab qiladi. Shu sababli train jarayonining barqarorligi (stability) eng birinchi dolzarb masalalardan.

Problem

Exploding Gradients

- Ko'p qavatli (100+ layers)
- Warm up qilinmaganida
- Loss / noisy data

Symptom

Loss birdaniga Dortlaydi

Loss = NaN yoki Loss = inf

Solution → LR Warmup (0 → Peak LR davomida)

Gradient Clipping (Norm threshold = 1.0)

Result

Loss Barqaror pasayishda tekis

Barqaror va so'nib boradi.

Exploding gradients.

1. Layers 100+
2. Boshlangich da loss ehtiyotkorlik bilan berilgan Warm up Qilinmaganida.
3. Traindata buzilgan / noisy yoki false.

Model butunlay ishdan chiqadi.

1. LR → Training bilan yuqori LR berganda dastlabki 2000 qadamgacha LR qiymatini 0 → LR max shkalasida chiziqli oshirib bor.
2. Gradien Clipping. Threshold 1.0

Regularization : Overfitting va LLM lardagi oltin qoida.

Overfitting : Model Datadagi umumiy mantiq va qonuniyatlarni o'rniga matnlarni ko'r-ko'rona yoddab oladi

Regularization : Model yoddab olish o'rniga matnning ichki semantik mantig'ini tushunishga majbur qiluvchi matematik usullar to'plami.

Regularization in LLM: Golden Rule

1. Weight Decay : Neyron Weight haddan tashqari osib ketsa, ushbu penalti uni orqaga tortadi va neyronlarni vazmin ushlab turadi.
2. Drop out : Vaqtincha baza Neyronlar o'chirish.

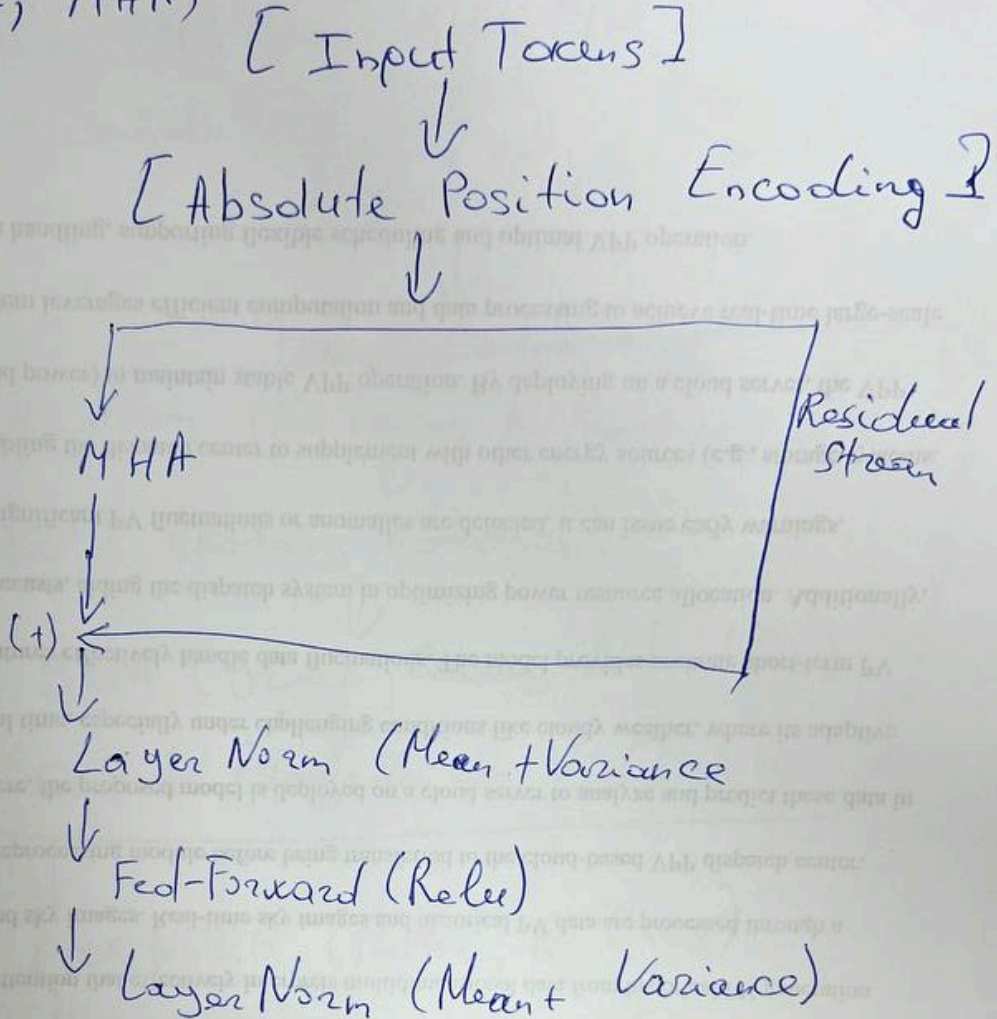
LLM da oltin qoida: LLM larni training qilishda Drop out = 0.0 qo'yiladi. Chunki trillionlab tokenlar ulkan Datasetda Drop out qo'llash underfitting va ortiqcha compute resurslarning isrof bo'lishiga olib keladi. Faqat Weight Decay ishlatilishi yetarlidir.

LLM Architecture

Eski (orginal Vaswani 2017)
Yangi (zamonaviy SOTA - Llama-3, Mistral)

Komponentlar	Eski xolat	Yangi xolat
Arxitektura turi	Encoder-Decoder	Decoder only
Normalizatsiya tartibi	Post-Norm (Amallardan song)	Pre-Norm (Amallardan oldin
Normalizatsiya Algoritmi	Layer Norm (Mean va Variance)	RMSNorm (10-50%)
Position Encoding	Absolute Sinusodial Positional Encoding	RoPE (Rotary Position Embedding Complex
Activation Function	Relu / gELU	SWiGLU High Reasoning
Attention Variant	MHA (Heavy KV)	GQA (Tejamkor
Blok tuzulishi	Ketma-ket	Serial yoki Kernel Fused Parallel
Dropout	$0.1 Model boylab	0.0

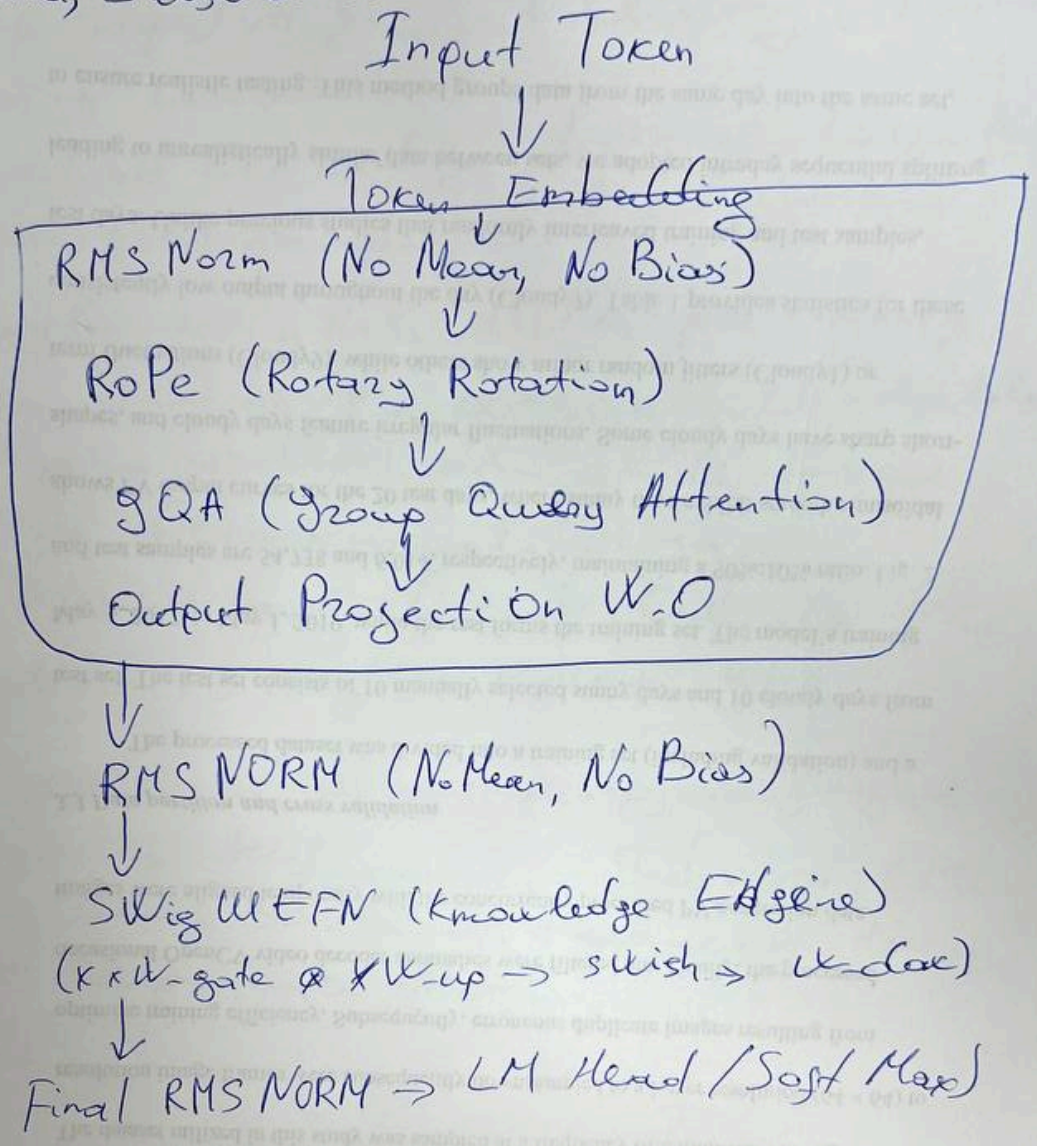
Esqi. Arxiterfuza (Post-Norm, Layer Norm, Relu, MHA)



Yangi arxitektura.

Pre-Norm, RMS Norm, SWiGLU, RoPe, GQA

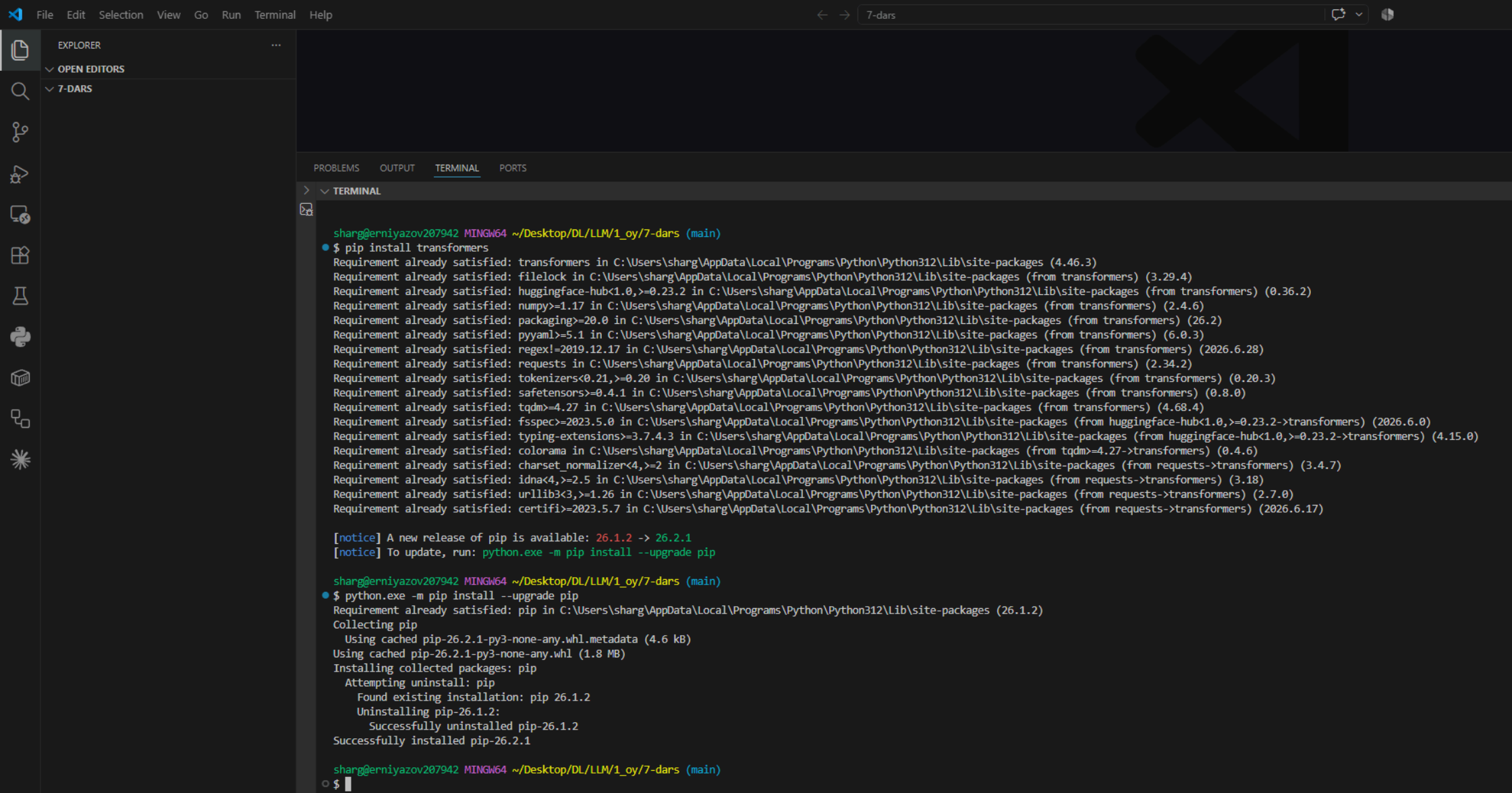
Llama-3 uslubida.

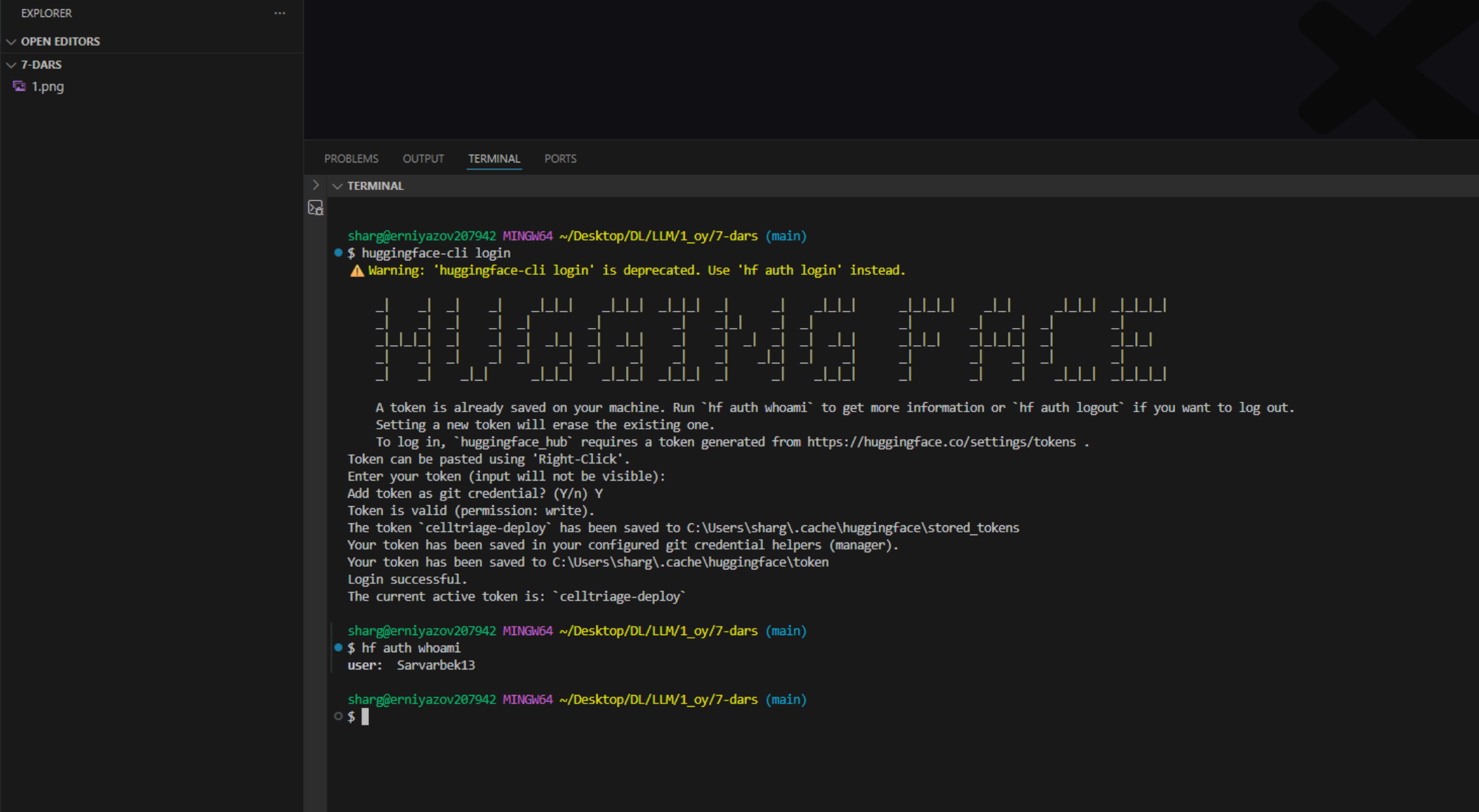


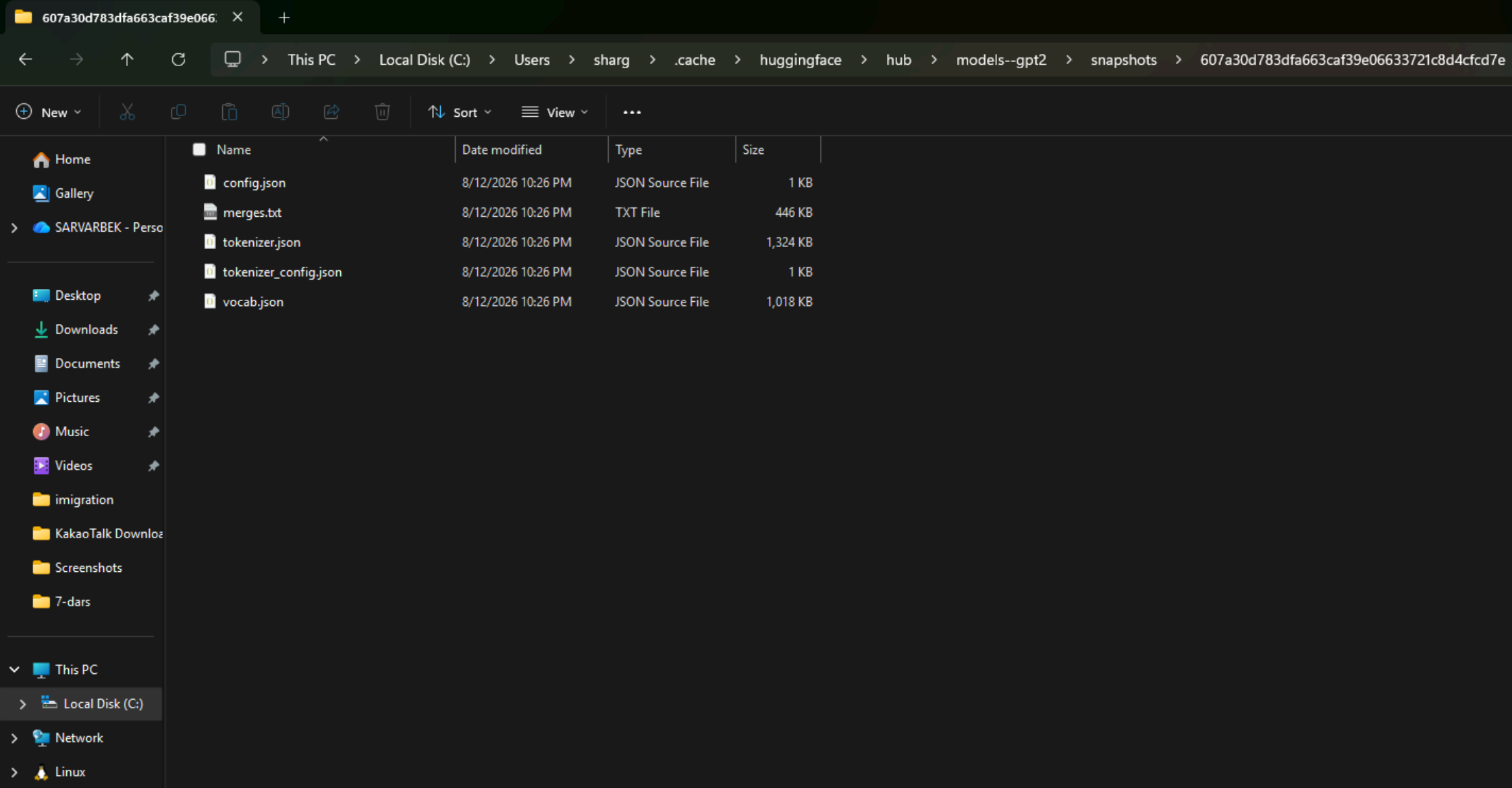
Xulosalar.

1. Training Optimizatsiya: LLM o'qitishda barqarorlikka erishish uchun LR Warm up (2000 steps) va Gradient Clipping (Threshold 1.0) hayotiy zaruratdir.
   Gradient Exploding bo'lganda Loss = NaN xatosi kelib chiqadi.
2. Regularizatsiya Rolu: Ulkan pretraining ma'lumotlarda Dropout = 0.0 qilib o'ditilgan ortiqcha yodlashlarni olidi faqat Weight Decay orqali olinadi.
3. Architecture → Decoder Only + Pre-Norm RmsNorm + SwiGLU + RoPE + GQA.
4. Tokenizatsiya Rolu: Kompyuter faqat raqamni tushunadi. Subword BPE algoritmi so'zlarni tegishli token ID raqamiga o'tkazadi.

Tokenizer matnni raqamga aylantiruvchi lug'at LLM modelining o'zi emas.
Ġ (U+0120) bo'sh joyni bildiruvchi maxsus belgi GPT-2 ning byte-level BPE usulida ishlatiladi.
Katta/kichik harf, tinish belgi, joylashuv, barchasi boshqa token beradi
Inglizcha matn samarali tokenlanadi
o'zbekcha matn esa 2 baravar ko'proq tokenga bo'linadi - sababi
Vocab asosan inglizcha ko'proq qismlardan iborat.


File Edit Selection View Go Run Terminal Help
7-dars
EXPLORER
OPEN EDITORS
7-DARS
PROBLEMS OUTPUT TERMINAL PORTS
TERMINAL
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ pip install transformers
Requirement already satisfied: transformers in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (4.46.3)
Requirement already satisfied: filelock in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (3.29.4)
Requirement already satisfied: huggingface-hub<1.0,>=0.23.2 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (0.36.2)
Requirement already satisfied: numpy>=1.17 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (2.4.6)
Requirement already satisfied: packaging>=20.0 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (2026.6.28)
Requirement already satisfied: requests in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (2.34.2)
Requirement already satisfied: tokenizers<0.21,>=0.20 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (0.20.3)
Requirement already satisfied: safetensors>=0.4.1 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.27 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from transformers) (4.68.4)
Requirement already satisfied: fsspec>=2023.5.0 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from huggingface-hub<1.0,>=0.23.2->transformers) (2026.6.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from huggingface-hub<1.0,>=0.23.2->transformers) (4.15.0)
Requirement already satisfied: colorama in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from tqdm>=4.27->transformers) (0.4.6)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from requests->transformers) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from requests->transformers) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from requests->transformers) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (from requests->transformers) (2026.6.17)
[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ python.exe -m pip install --upgrade pip
Requirement already satisfied: pip in C:\Users\sharg\AppData\Local\Programs\Python\Python312\Lib\site-packages (26.1.2)
Collecting pip
Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 26.1.2
Uninstalling pip-26.1.2:
Successfully uninstalled pip-26.1.2
Successfully installed pip-26.2.1
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$


EXPLORER
OPEN EDITORS
7-DARS
1.png
PROBLEMS
OUTPUT
TERMINAL
PORTS
TERMINAL
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ huggingface-cli login
⚠ Warning: 'huggingface-cli login' is deprecated. Use 'hf auth login' instead.
A token is already saved on your machine. Run `hf auth whoami` to get more information or `hf auth logout` if you want to log out.
Setting a new token will erase the existing one.
To log in, `huggingface_hub` requires a token generated from https://huggingface.co/settings/tokens .
Token can be pasted using 'Right-Click'.
Enter your token (input will not be visible):
Add token as git credential? (Y/n) Y
Token is valid (permission: write).
The token `celltriage-deploy` has been saved to C:\Users\sharg\.cache\huggingface\stored_tokens
Your token has been saved in your configured git credential helpers (manager).
Your token has been saved to C:\Users\sharg\.cache\huggingface\token
Login successful.
The current active token is: `celltriage-deploy`
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ hf auth whoami
user: Sarvarbek13
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$


607a30d783dfa663caf39e066
This PC
Local Disk (C:)
Users
sharg
.cache
huggingface
hub
models--gpt2
snapshots
607a30d783dfa663caf39e06633721c8d4cfcd7e
New
Sort
View
Home
Gallery
SARVARBEK - Perso
Desktop
Downloads
Documents
Pictures
Music
Videos
imigration
KakaoTalk Downloa
Screenshots
7-dars
This PC
Local Disk (C:)
Network
Linux
Name
Date modified
Type
Size
config.json 8/12/2026 10:26 PM JSON Source File 1 KB
merges.txt 8/12/2026 10:26 PM TXT File 446 KB
tokenizer.json 8/12/2026 10:26 PM JSON Source File 1,324 KB
tokenizer_config.json 8/12/2026 10:26 PM JSON Source File 1 KB
vocab.json 8/12/2026 10:26 PM JSON Source File 1,018 KB

File Edit Selection View Go Run Terminal Help

7-dars

EXPLORER

OPEN EDITORS

7-DARS

- 1.png
- 2.png
- 3.png
- tokenization.py

OUTLINE

TIMELINE

PROBLEMS OUTPUT TERMINAL PORTS

TERMINAL

```
sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ python tokenization.py
['I', 'Ġlove', 'ĠAI']
['Machine', 'Ġlearning', 'Ġis', 'Ġpowerful', '.']

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ cat >> tokenization.py << 'EOF'
print(tokenizer.tokenize("I LOVE ai"))
EOF
python tokenization.py
['I', 'Ġlove', 'ĠAI']
['Machine', 'Ġlearning', 'Ġis', 'Ġpowerful', '.']
['I', 'ĠLOVE', 'Ġa', 'i']

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ cat >> tokenization.py << 'EOF'
print(tokenizer.tokenize("Hello, world! How are you?"))
EOF
python tokenization.py
['I', 'Ġlove', 'ĠAI']
['Machine', 'Ġlearning', 'Ġis', 'Ġpowerful', '.']
['I', 'ĠLOVE', 'Ġa', 'i']
['Hello', ',', 'Ġworld', '!', 'ĠHow', 'Ġare', 'Ġyou', '?']

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ cat >> tokenization.py << 'EOF'
print(tokenizer.tokenize("I have 25 apples and 3 bananas."))
EOF
python tokenization.py
['I', 'Ġlove', 'ĠAI']
['Machine', 'Ġlearning', 'Ġis', 'Ġpowerful', '.']
['I', 'ĠLOVE', 'Ġa', 'i']
['Hello', ',', 'Ġworld', '!', 'ĠHow', 'Ġare', 'Ġyou', '?']
['I', 'Ġhave', 'Ġ25', 'Ġapples', 'Ġand', 'Ġ3', 'Ġbananas', '.']

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ cat >> tokenization.py << 'EOF'
print(tokenizer.tokenize("Bugun 7-darsni tugatdik"))
EOF
python tokenization.py
['I', 'Ġlove', 'ĠAI']
['Machine', 'Ġlearning', 'Ġis', 'Ġpowerful', '.']
['I', 'ĠLOVE', 'Ġa', 'i']
['Hello', ',', 'Ġworld', '!', 'ĠHow', 'Ġare', 'Ġyou', '?']
['I', 'Ġhave', 'Ġ25', 'Ġapples', 'Ġand', 'Ġ3', 'Ġbananas', '.']
['Bug', 'un', 'Ġ7', '-', 'd', 'ars', 'ni', 'Ġtug', 'at', 'd', 'ik']

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$ [200~cat >> tokenization.py << 'EOF'
> tokens = tokenizer.tokenize("I love AI")
> for t in tokens:
>     print(repr(t), "->", repr(tokenizer.convert_tokens_to_string([t])))
> EOF
bash: [200~cat: command not found

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
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for t in tokens:
    print(repr(t), "->", repr(tokenizer.convert_tokens_to_string([t])))
EOF
python tokenization.py
['I', 'Ġlove', 'ĠAI']
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['I', 'ĠLOVE', 'Ġa', 'i']
['Hello', ',', 'Ġworld', '!', 'ĠHow', 'Ġare', 'Ġyou', '?']
['I', 'Ġhave', 'Ġ25', 'Ġapples', 'Ġand', 'Ġ3', 'Ġbananas', '.']
['Bug', 'un', 'Ġ7', '-', 'd', 'ars', 'ni', 'Ġtug', 'at', 'd', 'ik']
'I' -> 'I'
'Ġlove' -> ' love'
'ĠAI' -> ' AI'

sharg@erniyazov207942 MINGW64 ~/Desktop/DL/LLM/1_oy/7-dars (main)
$
```

0 0